

DIFFERENTIAL EQUATIONS LECTURE NOTES

WEEK (VI)

CASE (III) REPEATED EIGENVALUES (algebraic multiplicity $\lambda > 1$) $\text{alg}(\lambda) > 1$

$$x' = Ax \quad A \text{ constant}$$

$$\det(A - \lambda I) = 0 \rightarrow \text{characteristic equation}$$

Suppose λ is an eigenvalue of A , with $\text{alg}(\lambda) = k > 1$
There are two possibilities

$$\sqrt{m(\lambda) = \text{alg}(\lambda)}$$

① k linearly independent $\text{eigenvectors } \lambda: \vec{v}_1, \vec{v}_2, \dots, \vec{v}_k$

The solutions

$$\begin{aligned} \vec{x}_1(t) &= e^{\lambda t} \vec{v}_1 \\ \vec{x}_2(t) &= e^{\lambda t} \vec{v}_2 \\ &\vdots \\ \vec{x}_k(t) &= e^{\lambda t} \vec{v}_k \end{aligned}$$

$$m(\lambda) = \text{alg}(\lambda)$$

$$\sqrt{m(\lambda) < \text{alg}(\lambda)} \quad \text{?}$$

② There are less than k linearly independent eigenvectors
 $m(\lambda) < \text{alg}(\lambda) = k$

of linearly independent eigenvectors

of free variables in the system

$$(A - \lambda I) \vec{v} = \vec{0}$$

$\text{alg}(\lambda) = \#$ an eigenvalue appears as the root of $\det(A - \lambda I) = 0$

Question: How can we find the missing eigenvectors which are generalized eigenvectors

$m(\lambda) = \#$ linearly independent eigenvectors

Definition To find eigenvectors we solve $(A - \lambda I) \vec{v} = \vec{0}$, actually we are looking for the kernel of the matrix $A - \lambda I$

$$\text{Ker}(A - \lambda I) = \{ \vec{v} \mid (A - \lambda I) \vec{v} = \vec{0} \}$$

= set of linearly independent eigenvectors corresponding to λ of matrix A .

$$\dim(\text{Ker}(A - \lambda I)) = \# \text{ of linearly independent eigenvectors} \rightarrow$$

$$= m(\lambda)$$

$$= \# \text{ of free variables in } (A - \lambda I) \vec{v} = \vec{0}$$

Ex Consider the case $\det(A - \lambda I) = m(\lambda)$

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \vec{x}. \text{ Solve the system}$$

Eigenvalues $\det(A - \lambda I) = 0$

$$\begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = -(\lambda+1)^2(\lambda-2) = 0 \Rightarrow$$

$$\lambda_1 = 2$$

$$\lambda_{2,3} = -1$$

repeated

$$m(2)$$

$$q.p(2) = 1$$

$$q.p(-1) = 2$$

Eigenvectors

$$\lambda = 2 \quad (A - 2I)\vec{v} = 0 \quad (\text{or Find } \ker(A - 2I))$$

$$\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \xrightarrow{\text{Row op.}} \begin{bmatrix} 1 & -1/2 & 1/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

v_1, v_2 : leading
 v_3 : free

$$m(2) = 1 = q.p(2)$$

$$v_1 - 1/2 v_2 + 1/2 v_3 = 0 \Rightarrow v_1 = 0$$

$$v_2 - v_3 = 0 \Rightarrow v_2 = v_3$$

$$\vec{v} = \begin{bmatrix} 0 \\ v_3 \\ v_2 \end{bmatrix} = v_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\ker(A - 2I) = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\} \Rightarrow \vec{x}_1(t) = e^{2t} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\lambda = -1 \quad (A + I)\vec{u} = 0 \quad (\ker(A + I)?)$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

u_1 : leading

u_2, u_3 : free $m(-1) = 2 = q.p(-1)$

$$u_1 + u_2 + u_3 = 0 \Rightarrow u_1 = -u_2 - u_3$$

$$\vec{u} = \begin{bmatrix} -u_2 - u_3 \\ u_2 \\ u_3 \end{bmatrix} = u_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + u_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\ker(A + I) = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\vec{x}_2(t) = e^{-t} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\vec{x}_3(t) = e^{-t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\vec{x}_{\text{part}}(t) = c_1 \vec{x}_1(t) + c_2 \vec{x}_2(t) + c_3 \vec{x}_3(t)$$

$$\vec{x}_{\text{part}}(t) = c_1 e^{2t} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c_3 e^{-t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

That is, A is diagonalizable $\exists$ an invertible matrix

$$P = \begin{bmatrix} 0 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$\downarrow$ $\downarrow$ $\downarrow$
 $\vec{v}_1$ $\vec{v}_2(t)$ $\vec{v}_3(t)$

such that $P^{-1}AP = J$ is diagonal: $J = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

Thm A matrix A is diagonalizable iff all the eigenvalues are real and every eigenvalue of $\text{alg}(\lambda) = k$ produces k linearly independent eigenvectors (i.e. $\text{alg}(\lambda) = m(\lambda)$)

Question What happens if A is not diagonalizable? In other words how can we find k linearly independent vectors for an eigenvalue λ with $m(\lambda) < \text{alg}(\lambda)$?

(Answer! we need to look Jordan forms)

Remember If $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ are fundamental solutions to homogeneous system $\vec{x}' = A\vec{x}$, the matrix

$$\overline{X} = [\vec{x}_1 \quad \vec{x}_2 \quad \dots \quad \vec{x}_n]$$

is called a fundamental matrix for $\vec{x}' = A\vec{x}$. Then, the general solution for (1) is

$$\vec{x}_p = \overline{X} \cdot C = c_1 \vec{x}_1 + c_2 \vec{x}_2 + \dots + c_n \vec{x}_n$$

moreover $\det \overline{X} = \chi(\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n) \neq 0 \Rightarrow \overline{X}$ is invertible

Then, one can easily solve I.V.P. $\begin{cases} \vec{x}' = A\vec{x} \\ \vec{x}(t_0) = \vec{x}_0 \end{cases}$

$$\Rightarrow \vec{x} = \bar{X} \cdot C \Rightarrow \text{put } X(t_0) = \bar{X}(t_0) \cdot C = \vec{x}_0 \Rightarrow \vec{C} = \bar{X}^{-1}(t_0) \cdot \vec{x}_0$$

Since $\bar{X}(t_0)$ is invertible

Then, the solution $\boxed{\vec{x}(t) = \bar{X}(t) \cdot \bar{X}^{-1}(t_0) \cdot \vec{x}_0}$

Thm If $\bar{X}(t)$ is a fundamental matrix for (H) $\vec{x}' = A\vec{x}$ then $\bar{X}(t) \cdot P$ is also a fundamental matrix of (H) for any nonsingular (invertible) matrix P ($\det P \neq 0$)

Proof Let $P = \bar{X}^{-1}(t_0) \Rightarrow \bar{X}(t) \cdot P = \bar{X}(t) \cdot \bar{X}^{-1}(t_0) = \Phi(t)$

such that $\Phi(t_0) = \bar{X}(t_0) \bar{X}^{-1}(t_0) = I$ (identity matrix)

That is, $\Phi(t) = [\vec{y}_1(t) \ \vec{y}_2(t) \ \dots \ \vec{y}_n(t)]$ where each column

Then the solution for I.V.P.

$$y_j^2(t_0) = e_j^2 = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \xrightarrow{j\text{th row}} \begin{cases} \vec{x}' = A\vec{x} \\ \vec{x}(t_0) = \vec{x}_0 \end{cases} \text{ becomes } \vec{x}(t) = \Phi(t) \vec{x}_0$$

$\parallel$
 $\bar{X}(t) \cdot \bar{X}^{-1}(t_0)$

Remark We usually take $t_0 = 0$.

Definition $\Phi(t) = \bar{X}(t) \bar{X}^{-1}(t_0)$ with $\bar{X}(t_0) = I$ is called

the ~~fundamental~~ ^{fundamental} ~~matrix~~ ^{matrix}

for (H) and moreover $\Phi(t)$ is

the unique. Here, $\bar{X}(t)$ is any fundamental matrix for (H)
fundamental matrix satisfying the condition $\Phi(t_0) = I$

Ex 7: $\vec{x}' = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \vec{x}$

Q Find $\underline{X}(t)$ and $\Phi(t)$ for $t=0$

Eigenvalues $\begin{vmatrix} 2-\lambda & -1 \\ 3 & -2-\lambda \end{vmatrix} = \lambda^2 - 1 = 0 \quad \lambda = \pm 1$ real, distinct eigenvalues

Eigenvectors $\lambda = 1 \quad \begin{bmatrix} 1 & -1 \\ 3 & -3 \end{bmatrix} \xrightarrow{R_2 \rightarrow -3R_1 + R_2} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$ v_1 : leading
 v_2 : free
 $v_1 - v_2 = 0$
 $v_1 = v_2$

$\text{Ker}(A - \lambda I) = \{ \vec{v} = (v_1, v_2) : v_1 - v_2 = 0 \} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$

$\vec{x}_1(t) = e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda = -1 \quad \begin{bmatrix} 3 & -1 \\ 3 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & -1 \\ 0 & 0 \end{bmatrix}$ v_1 : leading
 v_2 : free
 $3v_1 - v_2 = 0$

$\text{Ker}(A + I) = \left\{ \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} : 3v_1 - v_2 = 0 \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$

$\vec{x}_2(t) = e^{-t} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$\underline{X}(t) = [\vec{x}_1 \quad \vec{x}_2] = \begin{bmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix}$ any fundamental matrix

Find the special fundamental matrix $\Phi(t) = \underline{X}(t) \cdot \underline{X}^{-1}(t_0) = \underline{X}(t) \cdot \underline{X}^{-1}(0)$

Find $\underline{X}(0) \Rightarrow \underline{X}^{-1}(0)$

$\underline{X}(0) = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix} \Rightarrow \underline{X}^{-1}(0) = \frac{1}{2} \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix}$

Then, $\Phi(t) = \underline{X}(t) \cdot \underline{X}^{-1}(0) = \begin{bmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix} \cdot \frac{1}{2} \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix}$

$\Phi(t) = \frac{1}{2} \begin{bmatrix} 3e^t - e^{-t} & -e^t + e^{-t} \\ 3e^t - 3e^{-t} & -e^t + 3e^{-t} \end{bmatrix}$

Check $t=0 \quad \Phi(0) = I = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

⑥ Solve I.V.P $\vec{x}' = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \vec{x}$, $\vec{x}(0) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

$$\vec{x}_p = \Phi(t) \cdot \vec{c} \Rightarrow \vec{x}_p(0) = \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \underbrace{\Phi(0)}_{\mathbf{I}} \cdot \vec{c} \Rightarrow \vec{c} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

So, solution of I.V.P $\vec{x}(t) = \Phi(t) \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

On the other hand $\vec{x}_g = \vec{X}(t) \cdot \vec{c}$

$$= \vec{X}(0) \cdot \vec{c} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix} \cdot \vec{c} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

to find $\vec{c}$ we need to solve a system of equations.

Question - What is the meaning of $\Phi(t)$

It is the solution of I.V.P: $\begin{cases} \Phi'(t) = A \cdot \Phi(t) \\ \Phi(0) = I \end{cases}$

Now look at e^{At} (A is a matrix)

Matrix exponential e^{At} is called matrix exponential where A is a constant matrix

Remember $e^{at} = \sum_{k=0}^{\infty} \frac{(at)^k}{k!} = 1 + at + \frac{(at)^2}{2!} + \dots$ converges $\forall t$.

$$e^{At} = \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \dots \text{converges } \forall t.$$

$$= I + \sum_{k=1}^{\infty} \frac{(At)^k}{k!}$$

Check $(e^{At})' = \frac{d}{dt} (e^{At}) = \sum_{k=1}^{\infty} A \cdot k \frac{(At)^{k-1}}{k!} = A \sum_{k=1}^{\infty} \frac{(At)^{k-1}}{(k-1)!}$

$$= A \left(\sum_{k=0}^{\infty} \frac{(At)^k}{k!} \right) e^{At}.$$

(1)

$$\frac{d}{dt}(e^{At}) = A \cdot e^{At} \Rightarrow e^{At} \text{ satisfies (1) } \vec{x}' = A\vec{x}$$

Moreover, $e^{At}|_{t=0} = I$.

Thus, we can conclude that $e^{At} = \Phi(t)$ where $\Phi(t)$ is the special fundamental matrix for (1) $\vec{x}' = A\vec{x}$ with $\Phi(0) = I$

Definition The exponential matrix e^{At} where A is constant is the special ^{fundamental} matrix for homogeneous system

$$\vec{x}' = A\vec{x} \text{ which satisfies } e^{At}|_{t=0} = I.$$

That is, $e^{At} = \Phi(t) = \underline{X}(t) \underline{X}^{-1}(0)$ where $\underline{X}(t)$ is any fundamental matrix

Remark $e^{-At} = ?$

$$\begin{aligned} e^{-At} &= \Phi^{-1}(t) = (\underline{X}(t) \cdot \underline{X}^{-1}(0))^{-1} = (\underline{X}^{-1}(0))^{-1} \underline{X}^{-1}(t) \\ &= \underline{X}(0) \underline{X}^{-1}(t) \end{aligned}$$

Exm $\vec{x}' = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} \vec{x}$ Find $\underline{X}(t), \Phi(t)$ $e^A = e^{At}|_{t=1}$

Eigenvalues $\begin{vmatrix} 1-\lambda & 1 \\ 4 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 - 4 = (\lambda-3)(\lambda+1) = 0$
 $\lambda = 3 \quad \lambda = -1$

Eigenvectors $\lambda = 3 \quad \begin{bmatrix} -2 & 1 \\ 4 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix}$ $-2v_1 + v_2 = 0$
 $\downarrow$ $\rightarrow$ free leading

$$\text{Ker}(A - 3I) = \left\{ \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} : -2v_1 + v_2 = 0 \right\} = \text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

$$\vec{x}_1(t) = e^{3t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\lambda = -1 \quad \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow 2v_1 + v_2 = 0 \quad \ker(A+I) = \text{span}\left\{ \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$$

$$\vec{x}_2(t) = e^{-t} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$\underline{X}(t) = \begin{bmatrix} e^{3t} & -e^{-t} \\ 2e^{3t} & 2e^{-t} \end{bmatrix} \quad \underline{X}(0) = \begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix}$$

$$\underline{X}^{-1}(0) = \frac{1}{4} \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}$$

$$\text{So, } \Phi(t) = e^{At} = \underline{X}(t) \cdot \underline{X}^{-1}(0) = \frac{1}{4} \begin{bmatrix} e^{3t} & -e^{-t} \\ 2e^{3t} & 2e^{-t} \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}$$

$$e^{At} = \Phi(t) = \frac{1}{4} \begin{bmatrix} 2e^{3t} + 2e^{-t} & e^{3t} - e^{-t} \\ 4e^{3t} - 4e^{-t} & 2e^{3t} + 2e^{-t} \end{bmatrix} \quad \text{Observe that } \Phi(0) = I$$

$$e^A = e^{At} \Big|_{t=1} = \frac{1}{4} \begin{bmatrix} 2e^3 + 2e^{-1} & e^3 - e^{-1} \\ 4e^3 - 4e^{-1} & 2e^3 + 2e^{-1} \end{bmatrix}$$

Connection Between eigenvectors and e^{At}

e^{At} is the special fundamental matrix for $\vec{x}' = A\vec{x}(t)$

How can we use e^{At} (not use $\Phi(t)$)

① $A_{n \times n}$ is diagonal $A = D = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{bmatrix} \quad D^n = \begin{bmatrix} \lambda_1^n & & 0 \\ & \lambda_2^n & \\ & & \ddots \\ & & & \lambda_n^n \end{bmatrix}$

$$e^{At} = \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = I + At + \frac{A^2 t^2}{2!} + \dots$$

$$= \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} + \begin{bmatrix} \lambda_1 t & & 0 \\ & \lambda_2 t & \\ 0 & & \ddots \\ & & & \lambda_n t \end{bmatrix} + \begin{bmatrix} \lambda_1^2 t^2 / 2! & & 0 \\ & \lambda_2^2 t^2 / 2! & \\ & & \ddots & \\ & & & \lambda_n^2 t^2 / 2! \end{bmatrix} + \dots$$

$$= \begin{bmatrix} 1 + \lambda_1 t + \lambda_1^2 t^2 / 2! + \lambda_1^3 t^3 / 3! + \dots & & 0 \\ & \ddots & \\ 0 & & 1 + \lambda_2 t + \lambda_2^2 t^2 / 2! + \dots & \\ & & & \ddots \\ 0 & & & & 1 + \lambda_n t + \lambda_n^2 t^2 / 2! + \dots \end{bmatrix}$$

(8)

$$= \begin{bmatrix} e^{\lambda_1 t} & & 0 \\ & e^{\lambda_2 t} & \\ 0 & & e^{\lambda_n t} \end{bmatrix}$$

$$e^{At} = \begin{bmatrix} e^{\lambda_1 t} & & 0 \\ & e^{\lambda_2 t} & \\ 0 & & e^{\lambda_n t} \end{bmatrix}$$

(11) If A is a diagonalizable matrix, then $\exists$ an invertible matrix P such that

$$P^{-1}AP = D = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ & & \ddots \\ 0 & & & \lambda_n \end{bmatrix} \Rightarrow A = P D P^{-1}$$

$$P P^{-1} P D P^{-1} = P D^2 P^{-1}$$

$$e^{At} = e^{P D P^{-1} t} = I + P D P^{-1} t + \frac{(P D P^{-1})^2 t^2}{2!} + \dots$$

$$= \frac{I}{P \cdot P^{-1}} + P D P^{-1} t + P D^2 P^{-1} \frac{t^2}{2!} + P D^3 P^{-1} \frac{t^3}{3!} + \dots$$

$$= P \left(I + D t + \frac{D^2 t^2}{2!} + \dots \right) P^{-1}$$

$$= P e^{Dt} P^{-1}$$

$$= P \begin{bmatrix} e^{\lambda_1 t} & & 0 \\ & e^{\lambda_2 t} & \\ & & \ddots \\ 0 & & & e^{\lambda_n t} \end{bmatrix} P^{-1}$$

For example (Previous Thm)

$$A = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} \text{ is diagonalizable with } P = \begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix},$$

$$P^{-1} = \frac{1}{4} \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix} \text{ such that } A = P D P^{-1} \text{ where } D = \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}$$

$$e^{At} = P e^{Dt} P^{-1} = \begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} e^{3t} & 0 \\ 0 & e^{-t} \end{bmatrix} \frac{1}{4} \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 2e^{3t} + 2e^{-t} & e^{3t} - e^{-t} \\ 4e^{3t} - 4e^{-t} & 2e^{3t} + 2e^{-t} \end{bmatrix}$$

Remark Thm If $\tilde{X}(t)$ is a fundamental matrix for $\vec{x}' = A\vec{x}$, then $\tilde{X}(t)P$ is also a fundamental matrix where P is invertible

$$e^{At} = P \cdot e^{Dt} P^{-1} \\ = \begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} e^{3t} & 0 \\ 0 & e^{-t} \end{bmatrix} \frac{1}{4} \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}$$

written
two
times
by
mistake.

$$= \frac{1}{4} \begin{bmatrix} 2e^{3t} + 2e^{-t} & e^{3t} - e^{-t} \\ 4e^{3t} - 4e^{-t} & 2e^{3t} + 2e^{-t} \end{bmatrix}$$

Remark Thm: If $\tilde{X}(t)$ is a fundamental matrix for $\vec{x}' = A\vec{x}$, then $\tilde{X}(t)P$ is also a fundamental matrix where P is invertible. $((\tilde{X}(t)P)') = \tilde{X}'P = A(\tilde{X}P)$ Solution: $\vec{x}' = A\vec{x}$ satisfies

$$e^{At} = e^{PDP^{-1}t} = P \cdot e^{Dt} P^{-1} \xrightarrow{\text{Multiply with } P \text{ from right}} e^{At} P = P \cdot e^{Dt} P^{-1} P = P e^{Dt}$$

If A is diagonalizable. is also a fundamental matrix

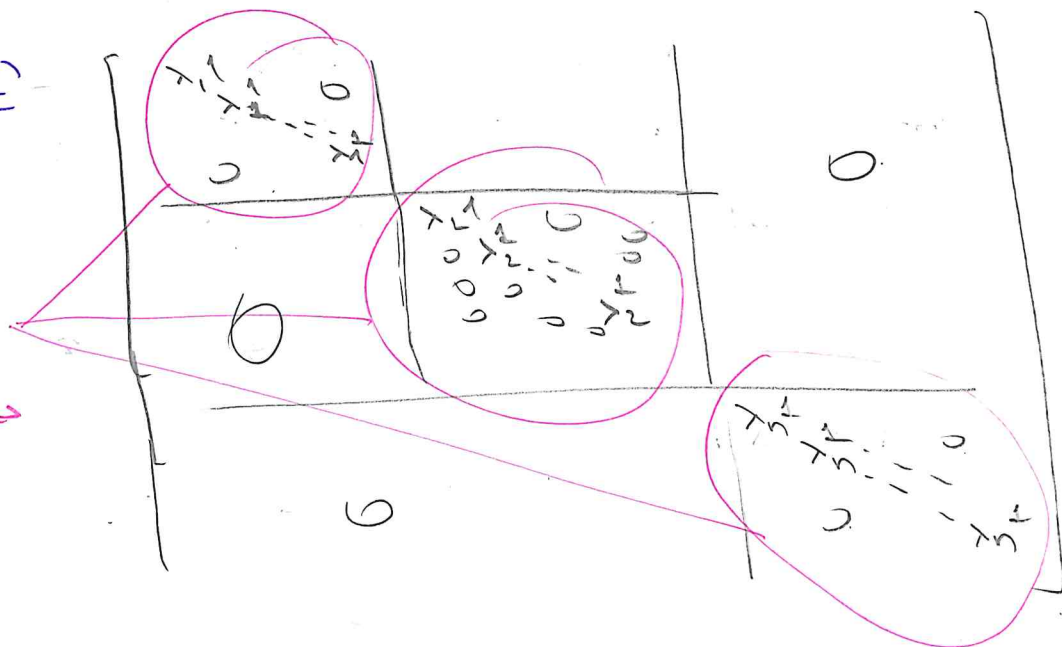
$$\text{So, } \boxed{\vec{x}_{particular} = P \cdot e^{Dt} \vec{c} \text{ for } (t)} \quad \text{for } (t)$$

Ex 111 If A is not diagonalizable, we need Jordan form

Thm For every matrix $A_{n \times n}$, there exists an invertible matrix P such that $P^{-1}AP = J$ where J is Jordan form.

Jordan Form

Jordan Blocks



$$J = \begin{bmatrix} J_1 & & 0 \\ & J_2 & \\ 0 & & \ddots \\ & & & J_n \end{bmatrix} \text{ where } J_i = \begin{bmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ 0 & & \ddots & 1 \\ & & & \lambda \end{bmatrix} \text{ or } J_i = \begin{bmatrix} \lambda & & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix}$$

Standard Jordan Canonical form

Ex 7

$$A = \begin{bmatrix} 5 & 4 & 2 & 1 \\ 0 & 1 & -1 & -1 \\ -1 & -1 & 3 & 0 \\ 1 & 1 & -1 & 2 \end{bmatrix}$$

$$\lambda = 1, 2, 4, 4$$

$$q.p.(1) = 1 = m(1)$$

$$q.p.(2) = 1 = m(2)$$

$$\textcircled{1} J_{1 \times 1} = [1]$$

$$\textcircled{2} J_{1 \times 1} = [2]$$

transpose version

$$m(4) = 1$$

$$J_{2 \times 2} = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$$

$$m(4) = 2$$

$$J = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$$

Remark

① $q.p.(x) = \text{size of Jordan form}$

② $m(x) = \# \text{ of Jordan blocks in a Jordan form}$

$$q.p.(4) = 2 \quad \begin{cases} m(4) = 1 = 1 \text{ Jordan block } J_{2 \times 2} = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix} \textcircled{3} \\ m(4) = 2 = 2 \text{ Jordan blocks } J_{2 \times 2} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \textcircled{4} \end{cases}$$

one eigenvector for $\lambda = 4$

$$J = \begin{bmatrix} 1 & & & \\ & 2 & & \\ & & 4 & 1 \\ & & 0 & 4 \end{bmatrix}$$

or

two eigenvectors for $\lambda = 4$

$$J = \begin{bmatrix} 1 & & & \\ & 2 & & \\ & & 4 & 0 \\ & & 0 & 4 \end{bmatrix}$$

Eigenvalue λ

$q.p.(x)$

$m(x)$

Jordan blocks

4

2

①

1 block, 2×2

$m(x) = 1$

2

1

1

1 block, 1×1

$m(x) = 2$

1

1

1

1 block, 1×1

each 1×1

total

2×2

$\textcircled{4}$

Ex $\det(A - \lambda I) = \lambda(\lambda - 1)^2(\lambda - 2) = 0.$

$$\lambda = 0 \Rightarrow \alpha_f(0) = 1 = n(0) \Rightarrow T_{1 \times 1} = \{0\}$$

$$\lambda = 2 \Rightarrow \text{q.f.}(\lambda = 2) = 1 = n(2) \rightarrow J_{1 \times 1} = \{2\}.$$

$$\lambda = 1 \Rightarrow \alpha_f(\lambda=1) = 2 \quad \text{---} \quad m(1) = 1 = \# \text{ of Jordan block } J_{2 \times 2} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

✓ $m(2) = 2 = \# \text{ of Jordan blocks } J_{2 \times 1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$J_{4 \times 4} = \begin{bmatrix} 0 & & & \\ & 2 & & \\ & & 1 & 1 \\ & & 0 & 1 \end{bmatrix} \quad \text{or} \quad J_{4 \times 4} = \begin{bmatrix} 0 & & & \\ & 2 & & \\ & & 1 & 0 \\ & & 0 & 1 \end{bmatrix}$$

Ex 7 $\det(A - \lambda I) = (\lambda - 1)^2(\lambda - 2)^3$

$$\lambda = 1 \quad \alpha_p(1) = 2 \quad - \quad m(1) = 1 \Rightarrow J_{2 \times 2} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$m(l+1)=2 \Rightarrow J_{2 \times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\lambda = 2 \quad \rho_p(2) = 3 \quad \underline{m(\lambda=2) = 1} \quad \rightarrow J_{3 \times 3} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$m(\lambda=2) = 2 \Rightarrow T_{3 \times 3} = \begin{pmatrix} 2 & & \\ & 2 & 1 \\ & 0 & 2 \end{pmatrix} \text{ or } T_{3 \times 3} = \begin{pmatrix} 2 & & \\ 0 & 2 & 1 \\ & & 2 \end{pmatrix}$$

$$m(\lambda=2) \rightarrow 3 \rightarrow J = \begin{bmatrix} 2 & & \\ & 2 & \\ & & 2 \end{bmatrix}$$

$$J_{5 \times 5} = \left[\begin{array}{cc|ccc} 1 & 1 & & & \\ 0 & 1 & & & \\ \hline & & 2 & 1 & 0 \\ & & 0 & 2 & 1 \\ & & 0 & 0 & 2 \end{array} \right] \text{ and the other}$$

Which one is true?

we need to find matrix P $P^{-1}AP = J$.

Question How can we find the invertible matrix P for the case of repeated eigenvalues.

Let's concentrate on only one Jordan block $J = \begin{bmatrix} \lambda & 1 & & 0 \\ & \lambda & 1 & \\ & & \ddots & \ddots \\ 0 & & & \lambda \end{bmatrix}_{n \times n}$

Find an invertible matrix P such that $P^{-1}AP = J$

$$P = [\vec{v}^{(1)} \quad \vec{v}^{(2)} \quad \dots \quad \vec{v}^{(n)}]$$

$$P^{-1}AP = J \Rightarrow AP = PJ$$

$$PA [\vec{v}^{(1)} \quad \vec{v}^{(2)} \quad \dots \quad \vec{v}^{(n)}] = [\vec{v}^{(1)} \quad \vec{v}^{(2)} \quad \dots \quad \vec{v}^{(n)}] \begin{bmatrix} \lambda & 1 & 0 & \dots & 0 \\ 0 & \lambda & 1 & \dots & 0 \\ & & \ddots & \ddots & \\ 0 & 0 & & \lambda & 1 \\ & & & & \lambda \end{bmatrix}$$

$$A\vec{v}^{(1)} = \lambda\vec{v}^{(1)}$$

$$A\vec{v}^{(2)} = \vec{v}^{(1)} + \lambda\vec{v}^{(2)}$$

$$A\vec{v}^{(3)} = \vec{v}^{(2)} + \lambda\vec{v}^{(3)}$$

$\vdots$

$$A\vec{v}^{(n)} = \vec{v}^{(n-1)} + \lambda\vec{v}^{(n)}$$

$$(A - \lambda I)\vec{v}^{(1)} = 0 \rightarrow \vec{v}^{(1)} \text{ is usual}$$

$$(A - \lambda I)\vec{v}^{(2)} = \vec{v}^{(1)} \quad \text{ordinary eigen}$$

$$(A - \lambda I)\vec{v}^{(3)} = \vec{v}^{(2)} \quad \vec{v}^{(2)}, \dots, \vec{v}^{(n)}$$

$$\vdots \quad \text{are called}$$

$$(A - \lambda I)\vec{v}^{(n)} = \vec{v}^{(n-1)} \quad \text{generalized}$$

eigenvectors which are obtained from eigenvector $\vec{v}^{(1)}$ by solving system

$$(A - \lambda I) / (A - \lambda I)\vec{v}^{(2)} = (A - \lambda I)\vec{v}^{(1)} \\ (A - \lambda I)^2 \vec{v}^{(2)} = (A - \lambda I)\vec{v}^{(1)} = 0$$

$$(A - \lambda I)\vec{v}^{(2)} = \vec{v}^{(1)} \rightarrow \text{solve for } \vec{v}^{(2)} \text{ with the obtained eigenvector } \vec{v}^{(1)} \text{ same system}$$

$$(A - \lambda I)\vec{v}^{(3)} = \vec{v}^{(2)} \Rightarrow (A - \lambda I)^2 \vec{v}^{(3)} = (A - \lambda I)\vec{v}^{(2)} = \vec{v}^{(1)}$$

multiply from left by $(A - \lambda I)$ multiply with $A - \lambda I$

$$(A - \lambda I)^3 \vec{v}^{(3)} = (A - \lambda I)\vec{v}^{(1)} = 0$$

This gives us a chain.

$$\underbrace{\text{Ker}(A-\lambda I)}_{\parallel} \subsetneq \underbrace{\text{Ker}(A-\lambda I)^2}_{\text{denote } V_{\lambda,2}} \subsetneq \underbrace{\text{Ker}(A-\lambda I)^3}_{\parallel} \subsetneq \dots \subsetneq \text{Ker}(A-\lambda I)^n$$

Choose $\vec{v}^{(1)} \in V_{\lambda,1} = \text{Ker}(A-\lambda I)$ but $\vec{v}^{(1)} \notin V_{\lambda,2}$

Choose $\vec{v}^{(2)} \in V_{\lambda,2} = \text{Ker}(A-\lambda I)^2$ but $\vec{v}^{(2)} \notin V_{\lambda,1}$

Choose $\vec{v}^{(3)} \in V_{\lambda,3}$ such that $\vec{v}^{(3)} \notin V_{\lambda,2}$

Finding J and P

First consider $A_{2 \times 2}$ and J has only one block

$$J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} \quad \text{Let } P = [\vec{v}^{(1)} \vec{v}^{(2)}]$$

$$P^{-1}AP = J \Rightarrow AP = PJ$$

$$A [\vec{v}^{(1)} \vec{v}^{(2)}] = [\vec{v}^{(1)} \vec{v}^{(2)}] \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$$

$$A\vec{v}^{(1)} = \lambda\vec{v}^{(1)} \Rightarrow \textcircled{1} (A-\lambda I)\vec{v}^{(1)} = 0 \quad \text{"ordinary eigenvector"}$$

$$A\vec{v}^{(2)} = \vec{v}^{(1)} + \lambda\vec{v}^{(2)} \Rightarrow \textcircled{2} (A-\lambda I)\vec{v}^{(2)} = \vec{v}^{(1)} \quad \text{"generalized eigenvector obtained by using } \vec{v}^{(1)} \text{"}$$

$$\textcircled{2} (A-\lambda I)\vec{v}^{(2)} = \vec{v}^{(1)} \xrightarrow{\text{multiply with } (A-\lambda I)} (A-\lambda I)^2\vec{v}^{(2)} = (A-\lambda I)\vec{v}^{(1)} = 0$$

$$(A-\lambda I)^2\vec{v}^{(2)} = 0 \Rightarrow \vec{v}^{(2)} \in \text{Ker}(A-\lambda I)^2 \quad \text{by } \textcircled{1}$$

and $\vec{v}^{(2)} \notin \text{Ker}(A-\lambda I)$

This produces a chain.

$$\underbrace{\text{Ker}(A-\lambda I)}_{\text{denote } V_{\lambda,1}} \subsetneq \underbrace{\text{Ker}(A-\lambda I)^2}_{\text{denote } V_{\lambda,2}}$$

denote $V_{\lambda,1}$

denote $V_{\lambda,2}$

Choose $\vec{v}^{(2)} \in V_{\lambda,2}$ but not in $V_{\lambda,1}$

Then, find $\vec{v}^{(1)}$ by using Equation $\textcircled{2}$

$$\vec{v}^{(1)} = (A-\lambda I)\vec{v}^{(2)}$$

$$\textcircled{2} \quad \vec{v}^{(1)} = (A - \lambda I) \vec{v}^{(2)}$$

↑ keep order

$$\text{Then } P = \begin{bmatrix} \vec{v}^{(1)} & \vec{v}^{(2)} \end{bmatrix} \Rightarrow J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$$

ordinary eigenvector generalized eigenvector

If we change order $P = [\vec{v}^{(2)} \quad \vec{v}^{(1)}] \Rightarrow J = \begin{bmatrix} \lambda & 0 \\ 1 & \lambda \end{bmatrix}$

To find $\vec{v}^{(1)}$ and $\vec{v}^{(2)}$ we have two ways

- ① First find $\vec{v}^{(2)}$, by choosing $\vec{v}^{(2)} \in \ker(A - \lambda I)^2$ but not in $\ker(A - \lambda I)$
 - ② First find $\vec{v}^{(1)}$, by finding $\ker(A - \lambda I)$ ($(A - \lambda I)\vec{v}^{(1)} = \vec{0}$)
- Then, find $\vec{v}^{(2)}$ by solving the system ②

Then, find $\vec{v}^{(2)}$ by solving the system ②

$$(A - \lambda I)\vec{v}^{(2)} = \vec{v}^{(1)} \quad \text{by substituting the found } \vec{v}^{(1)} \text{ on the R.H.S.}$$

Ex ① Find J and P for $A = \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix}$

Eigenvalues $(A - \lambda I) = \begin{vmatrix} 1-\lambda & -1 \\ 1 & 3-\lambda \end{vmatrix} = (\lambda - 2)^2 = 0 \Rightarrow \lambda_{1,2} = 2$
 $\text{alg}(\lambda = 2) = 2$

Eigenvectors $\ker(A - 2I) \quad ((A - 2I)\vec{v} = \vec{0})$

$$\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} v_1: \text{leading} \\ v_2: \text{free} \end{array} \quad \perp \text{ free } m(\lambda = 2) = 2 = \text{alg}(\lambda = 2)$$

$$\ker(A - 2I) = \left\{ \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} : v_1 + v_2 = 0 \right\} = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\} = v_{\lambda=2}$$

Then, $J_{2 \times 2} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$

To find P look for $\ker(A-2I)^2$ ($(A-2I)^2 \vec{v} = 0$)

$$(A-2I)^2 = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad 2 \text{ free variables}$$

$$\vec{v}^{(2)} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = v_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + v_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \ker(A-2I)^2 = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^2$$

Choose $\vec{v}^{(2)}$ such that $\vec{v}^{(2)} \in V_{\lambda,2}$ and $\vec{v}^{(2)} \notin V_{\lambda,1}$

Choose w.l.o.g. $\vec{v}^{(2)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then

$$\vec{v}^{(1)} = (A-2I)\vec{v}^{(2)} = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\text{Then, } P = \begin{bmatrix} \vec{v}^{(1)} & \vec{v}^{(2)} \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \quad P^{-1} = -1 \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\text{Check } J = P^{-1}AP = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

(b) Find solution of $\vec{x}' = \begin{bmatrix} 4 & -1 \\ 1 & 3 \end{bmatrix} \vec{x}$

$$e^{At} = e^{PJP^{-1}t} = P e^{Jt} P^{-1}$$

$$\vec{x}(t) = e^{At} \vec{p} = P e^{Jt} P^{-1} \vec{p} = P e^{Jt} \vec{c}$$

$$\begin{aligned} \vec{x} &= \vec{x}(t) \cdot \vec{c} = P e^{Jt} \vec{c} \\ &= \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} e^{Jt} \vec{c} \end{aligned}$$

What is e^{Jt} .

Thm (Exponentiating a matrix in Jordan Form)

If we consider one Jordan block $J = \begin{bmatrix} \lambda & 1 & 0 & \dots & 0 \\ 0 & \lambda & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & \lambda \end{bmatrix}_{n \times n}$

$$e^{Jt} = e^{\lambda t} \begin{bmatrix} 1 & t & \frac{t^2}{2!} & \frac{t^3}{3!} & \dots & \frac{t^{n-1}}{(n-1)!} \\ 0 & 1 & t & \frac{t^2}{2!} & \dots & \frac{t^{n-2}}{(n-2)!} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \frac{t^2}{2!} & \dots & 1 \end{bmatrix}_{n \times n}$$

If $J = \begin{bmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \\ & & & J_n \end{bmatrix} \Rightarrow e^{Jt} = \begin{bmatrix} e^{J_1 t} & & \\ & e^{J_2 t} & \\ & & \ddots \\ & & & e^{J_n t} \end{bmatrix}$

So back our example

$$J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \quad e^{Jt} = e^{2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

$$\begin{aligned} \vec{x}_{\text{general}} &= P \cdot e^{Jt} \vec{c} = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} e^{2t} & t e^{2t} \\ 0 & e^{2t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \\ &= \begin{bmatrix} -e^{2t} & -t e^{2t} + e^{2t} \\ e^{2t} & t e^{2t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \end{aligned}$$

$$= c_1 e^{2t} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} 1-t \\ t \end{bmatrix}$$

Solution comp from
ordinary eigenvector

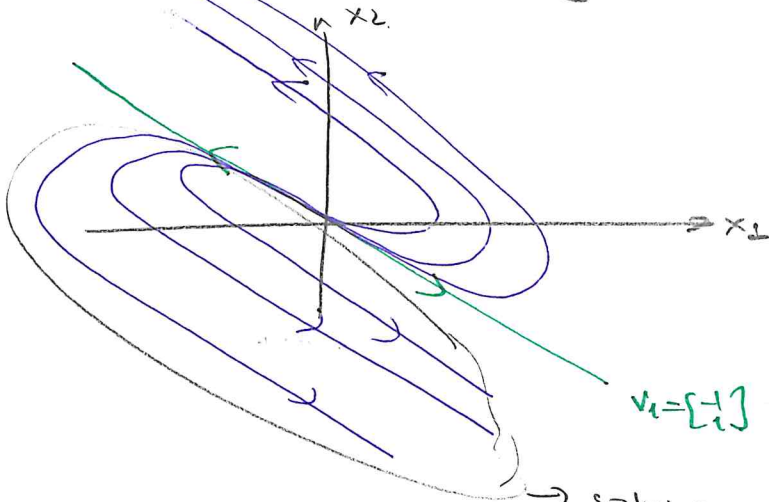
$$\vec{v}(1) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Solution comp
from generalized
eigenvector

$$\vec{v}(2) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Draw Phase Portrait

① Draw the ordinary eigenvector.



$$P = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \quad J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

If $\lambda > 0$ arrow will be outside

So origin is unstable since

all solutions go away

from origin as $t \rightarrow \infty$ $x(t) \rightarrow \infty$

If $\lambda < 0$ arrow will be towards origin so origin is stable

solutions in the other part of the eigenvector will be in opposite direction

② For the other solution pick a point

Let $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, then multiply with A $\vec{x}' = A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

Repeated Eigenvalues

~~Diagonalizable~~

SUMMARY

$$\vec{x}' = A \vec{x} \quad \vec{x} = e^{At} \vec{c} \quad e^{At} = \Phi(t) = X(t) \cdot X^{-1}(0)$$

If A is not diagonalizable: $\exists$ an invertible matrix P such that

$$A = P \cdot J \cdot P^{-1} \Rightarrow P^{-1} A P = J = \begin{matrix} \text{Jordan} \end{matrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

$$e^{At} = e^{PJP^{-1}t} = P \cdot e^{Jt} \cdot P^{-1}$$

$$e^{Jt} = e^{\lambda t} \begin{bmatrix} 1 & t & t^2/2! & \dots & t^{n-1}/(n-1)! \\ 0 & 1 & t & \dots & t^{n-2}/(n-2)! \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$$

$$J = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_n \end{bmatrix} \Rightarrow e^{Jt} = \begin{bmatrix} e^{J_1 t} & & & \\ & e^{J_2 t} & & \\ & & \ddots & \\ & & & e^{J_n t} \end{bmatrix}$$

Finding P and J

$A_{2 \times 2}$ Consider only Jordan Block $J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$, let $P = [\vec{v}^{(1)} \vec{v}^{(2)}]$

$$P^{-1}AP = J \Rightarrow AP = PJ$$

$$A [\vec{v}^{(1)} \vec{v}^{(2)}] = [\vec{v}^{(1)} \vec{v}^{(2)}] \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$$

$$A \vec{v}^{(1)} = \lambda \vec{v}^{(1)} \Rightarrow \textcircled{1} (A - \lambda I) \vec{v}^{(1)} = 0 \Rightarrow \vec{v}^{(1)} \in \ker(A - \lambda I)$$

"ordinary eigenvector"

$$A \vec{v}^{(2)} = \vec{v}^{(1)} + \lambda \vec{v}^{(2)} \Rightarrow \textcircled{2} (A - \lambda I) \vec{v}^{(2)} = \vec{v}^{(1)}$$

→ If we solve this system for $\vec{v}^{(2)}$ by using $\vec{v}^{(1)}$, $\vec{v}^{(2)}$ is called as "generalized eigenvector"

1st way Finding $\vec{v}^{(1)}, \vec{v}^{(2)}$

- ① Find $\vec{v}^{(1)}$ by looking at $\ker(A - \lambda I)$
- ② Use obtained $\vec{v}^{(1)}$ in ②, and solve the system $(A - \lambda I) \vec{v}^{(2)} = \vec{v}^{(1)}$ for $\vec{v}^{(2)}$

2nd way Finding $\vec{v}^{(1)}, \vec{v}^{(2)}$

- ① Find $\vec{v}^{(2)}$ by using ②

$$(A - \lambda I) \vec{v}^{(2)} = \vec{v}^{(1)} \quad \text{"let's multiply by } (A - \lambda I) \text{"}$$
$$(A - \lambda I)^2 \vec{v}^{(2)} = (A - \lambda I) \vec{v}^{(1)} = 0 \quad \text{by ①}$$
$$(A - \lambda I)^2 \vec{v}^{(2)} = 0 \Rightarrow \vec{v}^{(2)} \in \ker(A - \lambda I)^2 = V_{\lambda, 2}$$

So, choose $\vec{v}^{(2)} \in \ker(A - \lambda I)^2$ and $\vec{v}^{(2)} \notin \ker(A - \lambda I) = V_{\lambda, 1}$
to satisfy ②

guarantees that $\vec{v}^{(2)}$ and $\vec{v}^{(1)}$ are linearly independent

② Find $\vec{v}^{(1)}$ by using ② such that

$$\vec{v}^{(1)} = (A - \lambda I) \vec{v}^{(2)} \quad \text{matrix-vector multiplication}$$

there is no system solution.

After finding $\vec{v}^{(1)}$ and $\vec{v}^{(2)}$, just construct

$$P = \begin{bmatrix} \vec{v}^{(1)} & \vec{v}^{(2)} \end{bmatrix}$$

ordinary eigenvector
by using

$$\vec{v}^{(1)} = (A - \lambda I) \vec{v}^{(2)}$$

generalized eigenvector

Choose $\vec{v}^{(1)} \in \ker(A - \lambda I)^2$ and $\vec{v}^{(1)} \notin \ker(A - \lambda I)$

$$J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$$

$$\vec{x}_{\text{general}} = P \cdot e^{Jt} P^{-1} \vec{C}$$

$$\text{or } P e^{Jt} P^{-1}$$

$$\vec{x}_{\text{general}} = \underbrace{P e^{Jt}}_{\Sigma(t)} \vec{C}$$

since $\Sigma(t) = e^{At} P$ is also a fundamental matrix

Ex 7 Solve $\vec{x}' = \underbrace{\begin{bmatrix} -7 & 1 \\ -4 & -3 \end{bmatrix}}_A \vec{x}$ and draw phase portrait

Eigenvalues

$$\det(A - \lambda I) = \begin{vmatrix} -7-\lambda & 1 \\ -4 & -3-\lambda \end{vmatrix} \stackrel{\text{exercise}}{=} (\lambda + 5)^2 = 0 \Rightarrow \lambda_{1,2} = -5 \quad \text{alp}(-5) = 2$$

Size of Jordan form

Eigenvectors $\lambda = -5 \quad (A + 5I) \vec{v} = \vec{0} \quad (\text{Find } \ker(A + 5I))$

$$\begin{bmatrix} -2 & 1 \\ -4 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix}$$

v_1 : leading
 v_2 : free

|| # Jordan block)

1 free = 1 linearly independent

$$m(-5) = 1$$

$$J = \begin{bmatrix} -5 & 1 \\ 0 & -5 \end{bmatrix}$$

since $m(-5) = 1 < 2 = \text{alp}(-5)$

we have generalized eigenvector

$$\ker(A + 5I) = \left\{ \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} : -2v_1 + v_2 = 0 \right\}$$

$$-2v_1 + v_2 = 0 \Rightarrow v_2 = 2v_1$$

$$= \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

1st way to find eigenvectors

$$\vec{v}^{(1)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad (\text{w.l.o.g.}) \text{ ordinary eigenvector.}$$

Use (2) $(A + 5I)\vec{v}^{(2)} = \vec{v}^{(1)}$ and solve it for $\vec{v}^{(2)}$ by using

$$\begin{bmatrix} -2 & 1 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\vec{v}^{(1)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\vec{v}^{(2)} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} -2 & 1 & 1 \\ -4 & 2 & 2 \end{array} \right] \xrightarrow{-2R_1 + R_2 \Rightarrow R_1} \left[\begin{array}{cc|c} -2 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right] \Rightarrow -2v_1 + v_2 = 1$$

Let's choose
 $v_2 = 1, v_1 = 0$

$$\text{Then, } \vec{v}^{(2)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{Then: } P = [\vec{v}^{(1)} \ \vec{v}^{(2)}] = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \Rightarrow J = \begin{bmatrix} -5 & 1 \\ 0 & -5 \end{bmatrix}$$

$$\text{Check } J = P^{-1}AP = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} -7 & 1 \\ -4 & -3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -7 & 1 \\ 10 & -5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -5 & 1 \\ 0 & -5 \end{bmatrix}$$

Remark If we change order in P then J is changed

$$P = [\vec{v}^{(2)} \ \vec{v}^{(1)}] = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} \Rightarrow J = \begin{bmatrix} -5 & 0 \\ 1 & -5 \end{bmatrix}$$

2nd way to find eigenvectors

$$\ker(A + 5I) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

$$\vec{v}^{(1)} \in \ker(A + 5I) = \{ -2v_1 + v_2 = 0 \} = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

First find $\vec{v}^{(2)}$ by choosing $\vec{v}^{(2)} \in \ker(A + 5I)^2$ and $\vec{v}^{(2)} \notin \ker(A + 5I)$

$$\vec{v}^{(2)} \in \ker(A + 5I)^2 \quad (A + 5I)^2 = \begin{bmatrix} -2 & 1 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{2 free variables}$$

$\neq \ker(A + 5I)$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = v_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + v_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \ker(A + 5I)^2 = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^2 (\mathbb{C}^2)$$

Let's choose v.l.o.c. $\vec{v}^{(2)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, then find $\vec{v}^{(1)}$ by using

$$\vec{v}^{(1)} = (A + 5I)\vec{v}^{(2)} = \begin{bmatrix} -2 & 1 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \in \text{ker}(A + 5I)$$

Then $P = [\vec{v}^{(1)} \ \vec{v}^{(2)}] = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \quad J = \begin{bmatrix} -5 & 1 \\ 0 & -5 \end{bmatrix}$

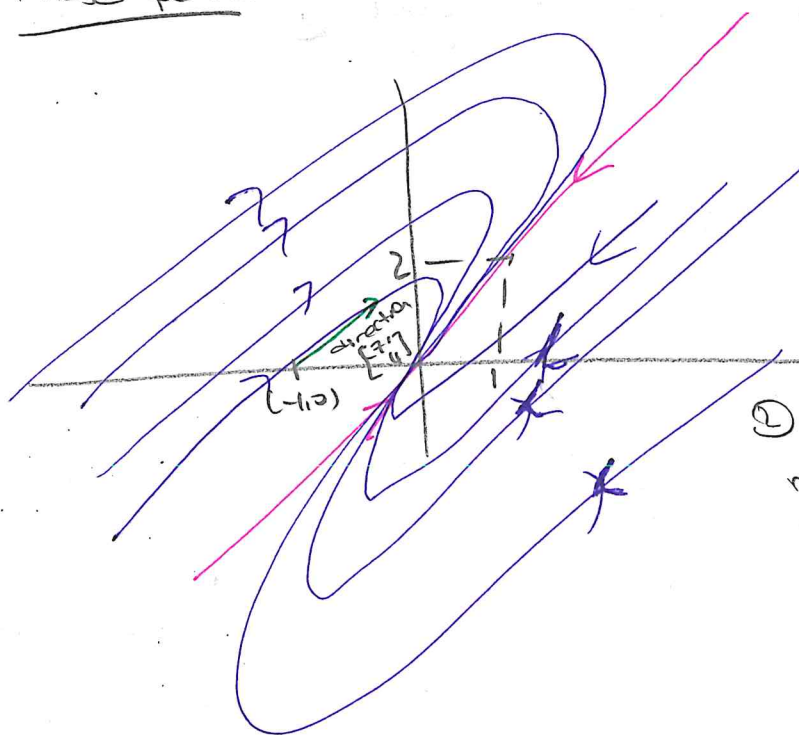
$$\vec{x}_{\text{general}} = P e^{Jt} \vec{c} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} e^{-5t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \vec{c}$$

$$(J = \begin{bmatrix} -5 & 1 \\ 0 & -5 \end{bmatrix}), e^{Jt} = e^{-5t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

$$\vec{x}_{\text{general}} = c_1 e^{-5t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 e^{-5t} \begin{bmatrix} t \\ 2t+1 \end{bmatrix} \quad t \rightarrow \infty \quad \vec{x}_{\text{general}} \rightarrow \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} t \\ 2t+1 \end{bmatrix}}_{\substack{\vec{v}^{(1)} \\ \vec{v}^{(2)}}} = t \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{\vec{v}^{(1)}} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\vec{v}^{(2)}}$$

Phase portrait



① Draw ordinary eigenvector $\vec{v}^{(1)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$\lambda = -5 < 0$ arrows will be towards the origin

Since all solutions goes origin as $t \rightarrow \infty$
origin is stable.

② Choose any point $\vec{x} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ and multiply with A

$$\begin{bmatrix} -7 & 1 \\ -4 & -3 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$$

A_{3x3}

Consider one Jordan Block

$$J = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$P = [\vec{v}^{(1)} \quad \vec{v}^{(2)} \quad \vec{v}^{(3)}]$$

$$P^{-1}AP = J \Rightarrow AP = PJ$$

$$A[\vec{v}^{(1)} \quad \vec{v}^{(2)} \quad \vec{v}^{(3)}] = [\vec{v}^{(1)} \quad \vec{v}^{(2)} \quad \vec{v}^{(3)}] \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A\vec{v}^{(1)} = \lambda\vec{v}^{(1)} \Rightarrow \textcircled{1} (A - \lambda I)\vec{v}^{(1)} = 0 \quad \vec{v}^{(1)} \in \ker(A - \lambda I) \text{ "ordinary vector"}$$

$$A\vec{v}^{(2)} = \vec{v}^{(1)} + \lambda\vec{v}^{(2)} \Rightarrow \textcircled{2} (A - \lambda I)\vec{v}^{(2)} = \vec{v}^{(1)} \Rightarrow \vec{v}^{(2)} \in \ker(A - \lambda I)^2$$
$$(A - \lambda I)^2 \vec{v}^{(2)} = (A - \lambda I)\vec{v}^{(1)} = 0 \quad \vec{v}^{(2)} \text{ "generalized eigenvector"}$$

by ①

$$A\vec{v}^{(3)} = \vec{v}^{(2)} + \lambda\vec{v}^{(3)} \Rightarrow \textcircled{3} (A - \lambda I)\vec{v}^{(3)} = \vec{v}^{(2)}$$
$$(A - \lambda I)^3 \vec{v}^{(3)} = (A - \lambda I)^2 \vec{v}^{(2)} = 0$$
$$\vec{v}^{(3)} \in \ker(A - \lambda I)^3 \quad \vec{v}^{(3)} \text{ is "generalized eigenvector"}$$

||
V_{λ,3}

So this produces the following chain

$$\underbrace{\ker(A - \lambda I)}_{= V_{\lambda,1}} \subsetneq \underbrace{\ker(A - \lambda I)^2}_{= V_{\lambda,2}} \subsetneq \underbrace{\ker(A - \lambda I)^3}_{= V_{\lambda,3} = \mathbb{R}^3}$$

i.e. $\vec{v}^{(1)} \in V_{\lambda,1}$ ordinary eigenvector

i.e. $\vec{v}^{(2)} \in V_{\lambda,2}$ and $\vec{v}^{(2)} \notin V_{\lambda,1}$

(choose $\vec{v}^{(3)}$ such that $\vec{v}^{(3)} \in \ker(A - \lambda I)^3 = V_{\lambda,3}$ and $\vec{v}^{(3)} \notin \ker(A - \lambda I)^2 = V_{\lambda,2}$)

$\vec{v}^{(2)}, \vec{v}^{(3)}$ "generalized eigenvectors"

Construct

$$\vec{v}^{(2)} = (A - \lambda I)\vec{v}^{(3)}$$
$$\vec{v}^{(1)} = (A - \lambda I)\vec{v}^{(2)}$$

Classification $A_{n \times n}$

Consider only one Jordan Block

$$J = \begin{bmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix}$$

$$P = [\vec{v}^{(1)} \quad \vec{v}^{(2)} \quad \dots \quad \vec{v}^{(n)}]$$

$AP = PJ$ will produce a chain

$$\ker(A - \lambda I) \subsetneq \ker(A - \lambda I)^2 \subsetneq \dots$$

$$\subsetneq \ker(A - \lambda I)^n (= \mathbb{R}^n(1))$$

Choose $\vec{v}^{(n)} \in \ker(A - \lambda I)^n = V_n$
and $\vec{v}^{(n)} \notin \ker(A - \lambda I)^{n-1} = V_{n-1}$

Then construct

$$\vec{v}^{(n-1)} = (A - \lambda I) \vec{v}^{(n)}$$

$$\vec{v}^{(n-2)} = (A - \lambda I) \vec{v}^{(n-1)}$$

$$\vdots$$

$$\vec{v}^{(1)}$$

$$= (A - \lambda I) \vec{v}^{(2)} \rightarrow \text{ordinary eigenvector}$$

} generalized eigenvector.

Ex 7

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -3 & 2 & 4 \end{bmatrix}$$

① Find P and J

② Find the solution of $\vec{x}' = A\vec{x}$

① Eigen values $\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 1 & 1 \\ 2 & 1-\lambda & -1 \\ -3 & 2 & 4-\lambda \end{vmatrix} \xrightarrow{\text{exercise}} -(1-\lambda)^3 = 0$
 $\lambda = 2$ $\text{alp}(2) = 3$

size of
Jordan
form

Eigenvectors $\lambda = 2$ $\ker(A - 2I) \quad ((A - 2I)\vec{v}^{(1)} = \vec{0})$

$$\begin{bmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ -3 & 2 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

v_1, v_2 : leading
 v_3 : free

1 free variable = 1 linearly independent eigenvector

$$J = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Jordan Block = $m(2) = 1$

$$\text{Ker}(A-2I) = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} : \begin{array}{l} -v_1 + v_2 + v_3 = 0 \\ v_2 + v_3 = 0 \end{array} \right\} \Rightarrow \left. \begin{array}{l} v_1 = 0 \\ v_2 = -v_3 \end{array} \right\}$$

↓
Choose $v_3 = 1 \quad v_2 = -1$

$$= \text{span} \left\{ \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\} = v_{\lambda=2,1} = v_{2,1}$$

Compute $v_{\lambda,2} = \text{Ker}(A-2I)^2$

$$(A-2I)^2 = \begin{bmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ -3 & 2 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ -3 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 1 & 1 \\ 1 & -1 & -1 \end{bmatrix} \xrightarrow[\text{operation}]{\text{row}} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

v_1 : leading

v_2, v_3 : free

$$v_1 = v_2 + v_3$$

$$\text{Ker}(A-2I)^2 = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} : v_1 - v_2 - v_3 = 0 \right\}$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} v_2 + v_3 \\ v_2 \\ v_3 \end{bmatrix} = v_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + v_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Ker}(A-2I)^2 = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\} \quad \text{Ker}(A-2I) \subsetneq \text{Ker}(A-2I)^2$$

$$\text{Ker}(A-2I)^3 = \mathbb{R}^3 = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + v_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + v_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\vec{v}(3) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$v_1 - v_2 - v_3 = -1 - 2 + 3 = 0$$

Choose $\vec{v}(2) = (A-2I)\vec{v}(3) = \begin{bmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ -3 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} \in \text{Ker}(A-2I)^2$
 $\notin \text{Ker}(A-2I)$

And $\vec{v}(1) = (A-2I)\vec{v}(2) = (A-2I)\begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \in \text{Ker}(A-2I)$
 † ordinary eigenvector

$$P = [\vec{v}(1) \quad \vec{v}(2) \quad \vec{v}(3)] = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & 0 \\ 1 & 3 & 0 \end{bmatrix} \Rightarrow J = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Check $P^{-1}AP$.

⑥ Find the solution of $\vec{x}' = A\vec{x}$

• $\vec{x}_{\text{general}} = \Sigma(t) \cdot \vec{C}$ where $\Sigma(t) = e^{At} \cdot P = P \cdot e^{Jt} \underbrace{P^{-1}P}_I = P \cdot e^{Jt}$

OR $\vec{x}_{\text{general}} = e^{At} \vec{C} = P \cdot e^{Jt} P^{-1} \vec{C}$

$$e^{Jt} = e^{2t} \begin{bmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} \vec{x}_{\text{general}} &= P \cdot e^{Jt} \vec{C} = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & 0 \\ 1 & -3 & 0 \end{bmatrix} e^{2t} \begin{bmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix} \vec{C} \\ &= e^{2t} \begin{bmatrix} 0 & -1 & -t+1 \\ -1 & -t+2 & -t^2/2+2t \\ 1 & t-3 & t^2/2-3t \end{bmatrix} \vec{C} \end{aligned}$$

$$\begin{aligned} \vec{x}_{\text{general}} &= C_1 \cdot e^{2t} \underbrace{\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}}_{\vec{v}(1)} + C_2 \cdot e^{2t} \underbrace{\begin{bmatrix} -1 \\ t+2 \\ t-3 \end{bmatrix}}_{\begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + t \underbrace{\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}}_{\vec{v}(1)}} + C_3 \cdot e^{2t} \underbrace{\begin{bmatrix} -t+1 \\ -t^2/2+2t \\ t^2/2-3t \end{bmatrix}}_{\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} + t \underbrace{\begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix}}_{\vec{v}(1)} + \frac{t^2}{2} \underbrace{\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}}_{\vec{v}(1)}} \end{aligned}$$

Ex Solve $\vec{x}' = A\vec{x}$ where $A = \begin{bmatrix} -3 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$

Eigenvalues $\det(A - \lambda I) = \begin{vmatrix} -3-\lambda & 2 & -1 \\ 0 & -3-\lambda & 0 \\ 0 & 0 & -3-\lambda \end{vmatrix} = (-3-\lambda)^3 = 0 \Rightarrow \lambda = -3$
 $\text{alg}(\lambda = -3) = 3$
 size of Jordan block

Eigenvectors $\lambda = -3$ $\text{Ker}(A + 3I)$

$$\begin{bmatrix} 0 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} v_2 \text{ is leading} \\ v_1, v_3 \text{ are free} \end{matrix} \quad 2 \text{ free} = m(-3) = 2 = \# \text{ of Jordan blocks}$$

$$J_{3 \times 3} = \left[\begin{array}{c|cc} -3 & 0 & 0 \\ 0 & -3 & 1 \\ 0 & 0 & -3 \end{array} \right] \quad \text{or} \quad J_{3 \times 3} = \left[\begin{array}{c|cc} -3 & 1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{array} \right]$$

$\text{Ker}(A + 3I) = \{ \vec{v} \mid 2v_2 - v_3 = 0 \} \quad v_2 = \frac{1}{2} v_3$ Let's choose

$v_3 = 1 \quad v_2 = \frac{1}{2}$ $\text{Ker}(A + 3I) = \text{span} \left\{ \begin{bmatrix} 1 \\ 1/2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1/2 \\ 1 \end{bmatrix} \right\}$

Since $\dim(\text{Ker}(A + 3I)) = 2$, we have two lin. independent eigenvectors. Hence we can (26) construct one Jordan chain of

Now we are sure that $v^{(3)}$ depends on $v^{(1)}$ or $v^{(2)}$ because we must have two linearly independent eigenvectors. And, depending on this information we will generate the Jordan form

$v^{(3)}$ generated by $v^{(1)}$

$$(A - \lambda I) \vec{v}^{(3)} = \vec{v}^{(1)}$$

$$(A - \lambda I)^2 \vec{v}^{(3)} = (A - \lambda I)^2 \vec{v}^{(1)} = \vec{0}$$

$$\vec{v}^{(3)} \in \ker(A - \lambda I)^2$$

$v^{(3)}$ generated by $v^{(2)}$

$$(A - \lambda I) \vec{v}^{(3)} = \vec{v}^{(2)}$$

$$(A - \lambda I)^2 \vec{v}^{(3)} = (A - \lambda I)^2 \vec{v}^{(2)} = \vec{0}$$

$$\vec{v}^{(3)} \in \ker(A - \lambda I)^2$$

For both cases $\vec{v}^{(3)} \in \ker(A - \lambda I)^2$, because we will have

Q structure

$$\begin{bmatrix} \vec{q} & \vec{v}^{(3)} \end{bmatrix} \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} = A \begin{bmatrix} \vec{q} & \vec{v}^{(3)} \end{bmatrix}$$

$$\vec{q} \lambda = A \vec{q} \Rightarrow (A - \lambda I) \vec{q} = \vec{0}$$

$$\vec{q} + \lambda \vec{v}^{(3)} = A \vec{v}^{(3)} \Rightarrow (A - \lambda I) \vec{v}^{(3)} = \vec{0}$$

$$(A - \lambda I)^2 \vec{v}^{(3)} = \vec{0}$$

our aim is to decide $\vec{q}$, $\vec{q}$ can be $\vec{v}^{(1)}$ or $\vec{v}^{(2)}$

$$(A + 3I)^2 = \begin{bmatrix} 0 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow 3 \text{ free variable}$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + v_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + v_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Choose $\vec{v}^{(3)} \in \ker(A + 3I)^2$ and $\vec{v}^{(3)} \notin \ker(A + 3I) \Rightarrow \vec{v}^{(3)} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$
 Now, we can choose $\vec{v}^{(2)} = (A + 3I) \vec{v}^{(3)}$ or $\vec{v}^{(2)} = (A + 3I)^2 \vec{v}^{(3)}$ we choose the first one

Then, $\vec{v}^{(2)} = (A + 3I) \vec{v}^{(3)} = \begin{bmatrix} 0 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \in \ker(A + 3I)$

(27)

$$\textcircled{*} \vec{v}^{(1)} = (A+3I) \vec{v}^{(2)} = \begin{bmatrix} 0 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \text{It can't be an eigenvector since it is zero vector}$$

$\vec{v}^{(1)}$ must be in $\ker(A+3I)$ and be linearly independent

from $\vec{v}^{(3)}$ and $\vec{v}^{(2)}$. Choose $\vec{v}^{(1)} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \in \ker(A+3I)$

$$\text{Then, } P = \begin{bmatrix} \underbrace{\vec{v}^{(1)}}_{\text{separate}} & \underbrace{\vec{v}^{(2)} \quad \vec{v}^{(3)}}_{\text{related eigenvectors}} \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 1 \\ 2 & 0 & 0 \end{bmatrix}$$

$$J = \begin{bmatrix} \underbrace{-3}_{J_1} & 0 & 0 \\ 0 & \underbrace{-3 \quad 1}_{J_2} \\ 0 & 0 & -3 \end{bmatrix}$$

$$e^{Jt} = \begin{bmatrix} e^{J_1 t} & & \\ & e^{J_2 t} & \\ & & e^{J_3 t} \end{bmatrix} = e^{-3t} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Then, } \vec{x}_{\text{general}} = P \cdot e^{Jt} \vec{c} = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 1 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} e^{-3t} & 0 & 0 \\ 0 & e^{-3t} & te^{-3t} \\ 0 & 0 & e^{-3t} \end{bmatrix} \vec{c}$$

$$= \begin{bmatrix} 0 & 2e^{-3t} & 2te^{-3t} \\ e^{-3t} & 0 & e^{-3t} \\ 2e^{-3t} & 0 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$= c_1 e^{-3t} \underbrace{\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}}_{\vec{v}^{(1)}} + c_2 e^{-3t} \underbrace{\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}}_{\vec{v}^{(2)}} + c_3 e^{-3t} \left(\underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{\vec{v}^{(3)}} + \underbrace{\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} t}_{\vec{v}^{(2)}} \right)$$

Solutions coming from ordinary eigenvector.

Solutions coming from generalized eigenvector.

Ex Find A^{219} if $A = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 3 & 6 & 2 \end{bmatrix}$

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 0 & 0 \\ -4 & 1-\lambda & 0 \\ 3 & 6 & 2-\lambda \end{vmatrix} = (1-\lambda)^2(2-\lambda) = 0 \Rightarrow \lambda_1 = 2, \lambda_{2,3} = 1$$

$\swarrow$
 $\text{alp}(2) = 1$
 $\text{m}(2) = 1$
 $J_1 = [2]_{1 \times 1}$

$\lambda_1 = 2$ $\text{Ker}(A - 2I)$

$$\begin{bmatrix} -1 & 0 & 0 \\ -4 & -1 & 0 \\ 3 & 6 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} v_1 = 0 \\ v_2 = 0 \\ v_3 = \text{free} \end{matrix} \quad \vec{v} = \begin{bmatrix} 0 \\ 0 \\ v_3 \end{bmatrix} + v_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Ker}(A - 2I) = \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \Rightarrow \vec{v} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$\lambda_{2,3} = 1$ $\text{Ker}(A - I)$

$$\begin{bmatrix} 0 & 0 & 0 \\ -4 & 0 & 0 \\ 3 & 6 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 6 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} v_1 = 0 \\ 6v_2 + v_3 = 0 \\ v_3 = \text{free} \end{matrix}$$

Size of
Jordan form

$$1 \text{ free} = \text{m}(1) = 1 < \text{alp}(1) = 2$$

1 Jordan Block

$$J_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{Ker}(A - I) = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} : v_1 = 0, v_3 = -6v_2 \right\}$$

$$= \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ -6 \end{bmatrix} \right\}$$

$$\text{Ker}(A - I)^2 = ?$$

leading

$$(A - I)^2 = \begin{bmatrix} 0 & 0 & 0 \\ -4 & 0 & 0 \\ 3 & 6 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ -4 & 0 & 0 \\ 3 & 6 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -21 & 6 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -21 & 6 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Ker}(A - I)^2 = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} : -21v_1 + 6v_2 + v_3 = 0 \right\} = \text{span} \left\{ \begin{bmatrix} 6/21 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1/21 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$v_2, v_3 = \text{free}$

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 6/21 v_2 + 1/21 v_3 \\ v_2 \\ v_3 \end{bmatrix} = v_2 \begin{bmatrix} 6/21 \\ 1 \\ 0 \end{bmatrix} + v_3 \begin{bmatrix} 1/21 \\ 0 \\ 1 \end{bmatrix}$$

(29)

Choose $\vec{v}^{(1)} \in \text{Ker}(A-I)^2$ and $\vec{v}^{(1)} \notin \text{Ker}(A-I)$, choose $\vec{v}^{(1)} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

then; $\vec{v}^{(2)} = (A-I)\vec{v}^{(1)} = \begin{bmatrix} 0 & 0 & 0 \\ -4 & 0 & 0 \\ 3 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ -4 \\ 24 \end{bmatrix}$

$P = \left[\begin{array}{c|c} \vec{v}^{(1)} & \vec{v}^{(2)} \\ \hline \vec{v} \end{array} \right] = \begin{bmatrix} 0 & 1 & 0 \\ -4 & 0 & 0 \\ 24 & 21 & 1 \end{bmatrix} \quad J = \left[\begin{array}{c|c} J_2 & \\ \hline & J_1 \end{array} \right]$

$\lambda=1$ $\lambda=2$

$$J = \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right]$$

or $P = \left[\vec{v} \quad \begin{bmatrix} \vec{v}^{(1)} & \vec{v}^{(2)} \end{bmatrix} \right] \quad J = \left[\begin{array}{c|c} J_1 & \\ \hline & J_2 \end{array} \right] = \left[\begin{array}{c|cc} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{array} \right]$

$A^{219} \approx J = P^{-1}AP$ $A = PJP^{-1}$
219 times

$A^{219} = \underbrace{PJP^{-1}}_I \cdot \underbrace{PJP^{-1}}_I \cdot \underbrace{PJP^{-1}}_I \cdot \dots \cdot \underbrace{PJP^{-1}}_I$

$A^{219} = P J^{219} P^{-1}$

$J^{219} = \left[\begin{array}{c|cc} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{array} \right]^{219}$

Remark (1) $J = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \mu & 0 \\ 0 & 0 & \theta \end{bmatrix}$

$J^n = \begin{bmatrix} \lambda^n & 0 & 0 \\ 0 & \mu^n & 0 \\ 0 & 0 & \theta^n \end{bmatrix}$

(2) $J = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$

$J^n = \begin{bmatrix} \lambda^n & n\lambda^{n-1} & \frac{n(n-1)}{2}\lambda^{n-2} \\ 0 & \lambda^n & n\lambda^{n-1} \\ 0 & 0 & \lambda^n \end{bmatrix}$

(3) $J = \left[\begin{array}{cc|c} \lambda & 1 & 0 \\ 0 & \lambda & 0 \\ \hline 0 & 0 & \mu \end{array} \right]$

$J^n = \left[\begin{array}{cc|c} \lambda^n & n\lambda^{n-1} & 0 \\ 0 & \lambda^n & 0 \\ \hline 0 & 0 & \mu^n \end{array} \right]$

$J^{219} = \left[\begin{array}{cc|cc} 2^{219} & 0 & 0 & 0 \\ 0 & 1 & 219 & 219 \\ \hline 0 & 0 & 0 & 1 \end{array} \right] = \begin{bmatrix} 2^{219} & 0 & 0 \\ 0 & 1 & 219 \\ 0 & 0 & 1 \end{bmatrix}$

$$A^{219} = P J^{219} P^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -4 & 0 \\ 1 & 24 & 21 \end{bmatrix} \begin{bmatrix} 2^{219} & 0 & 0 \\ 0 & 1 & 219 \\ 0 & 0 & 1 \end{bmatrix} P^{-1} \quad \text{Exercise}$$

HW

Ex Find e^{At} for $x' = Ax$ where $A = \begin{bmatrix} 5 & 0 & 4 \\ -6 & 1 & -1 \\ -6 & 0 & -5 \end{bmatrix}$

$$\det(A - \lambda I) = \begin{vmatrix} 5-\lambda & 0 & 4 \\ -6 & 1-\lambda & -1 \\ -6 & 0 & -5-\lambda \end{vmatrix} = (5-\lambda)(-1)^{1+1} \begin{vmatrix} 1-\lambda & -1 \\ 0 & -5-\lambda \end{vmatrix} + 4(-1)^{1+3} \begin{vmatrix} -6 & 1-\lambda \\ -6 & 0 \end{vmatrix}$$

$$= (5-\lambda)(1-\lambda)(-5-\lambda) + 4 \cdot 6(1-\lambda) = (1-\lambda)[(5-\lambda)(-5-\lambda) + 24]$$

$$= (1-\lambda)[-25 - 5\lambda + 5\lambda + 24 + \lambda^2] = (1-\lambda)(\lambda^2 - 1) = (1-\lambda)(\lambda-1)(\lambda+1)$$

$$\lambda_{1,2} = 1 \quad \lambda_3 = -1$$

Eigenvectors $\lambda_{1,2} = 1 \quad (A - \lambda I)\vec{v} = \vec{0}$

$$\begin{bmatrix} 4 & 0 & 4 \\ -6 & 0 & -1 \\ -6 & 0 & -6 \end{bmatrix} \xrightarrow{3/2 R_1 + R_3 \rightarrow R_3} \begin{bmatrix} 4 & 0 & 4 \\ -6 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{3/2 R_1 + R_2 \rightarrow R_2} \begin{bmatrix} 4 & 0 & 4 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1/4 \rightarrow R_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{leading}$$

$$\begin{cases} v_1 + v_3 = 0 \\ v_3 = 0 \end{cases} \quad v_1 = 0$$

v_2 is free one eigenvector

$$V_{\lambda=1} = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$\lambda_3 = -1 \quad (A + I)\vec{v} = \vec{0}$

$$\begin{bmatrix} 6 & 0 & 4 \\ -6 & 2 & -1 \\ -6 & 0 & -4 \end{bmatrix}$$

$V^{(3)}$

$$\xrightarrow{R_1 + R_2 \rightarrow R_2} \xrightarrow{R_1 + R_3 \rightarrow R_3} \begin{bmatrix} 6 & 0 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1/6 \rightarrow R_1} \xrightarrow{R_2/2} \begin{bmatrix} 1 & 0 & 2/3 \\ 0 & 1 & 3/2 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{leading}$$

Choose $v^{(3)} = \begin{bmatrix} -2/3 \\ -3/2 \\ 1 \end{bmatrix}$

$$v_1 + \frac{2v_3}{3} = 0$$

$$v_2 + 3/2 v_3 = 0$$

$$v_3 = \text{free}$$

$$v_{\lambda=-1} = \left\{ \begin{bmatrix} -2/3 \\ -3/2 \\ 1 \end{bmatrix} \right\}$$

$$v = \begin{bmatrix} -2/3 v_3 \\ -3/2 v_3 \\ v_3 \end{bmatrix}$$

Choose

$$v^{(3)} = \begin{bmatrix} -4 \\ -9 \\ 6 \end{bmatrix}$$

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Eigenvalues. $v^{(1)}$ and $v^{(2)}$, we will have

$$\begin{bmatrix} v^{(1)} & v^{(2)} \end{bmatrix} \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} = A \begin{bmatrix} v^{(1)} & v^{(2)} \end{bmatrix} \quad \lambda v^{(1)} = A v^{(1)} \Rightarrow v^{(1)}(A - \lambda I) = 0$$

$$v^{(1)} + \lambda v^{(2)} = A v^{(2)}$$

For our case $\lambda = 1$

$$(A - \lambda I)v^{(2)} = v^{(1)}$$

$$(A - \lambda I)^2 v^{(2)} = 0$$

So $v^{(1)} \in \ker(A - I)$, $v^{(2)} \in \ker(A - I)^2$

$$(A - I)^2 = \begin{bmatrix} 4 & 0 & 4 \\ -6 & 0 & -1 \\ -6 & 0 & -6 \end{bmatrix} \begin{bmatrix} 4 & 0 & 4 \\ -6 & 0 & -1 \\ -6 & 0 & -6 \end{bmatrix} = \begin{bmatrix} -8 & 0 & -8 \\ -18 & 0 & -18 \\ 12 & 0 & 12 \end{bmatrix}$$

$$\xrightarrow{R_1 \rightarrow -1/8 R_1} \begin{bmatrix} 1 & 0 & 1 \\ -18 & 0 & -18 \\ 12 & 0 & 12 \end{bmatrix} \xrightarrow{R_2 + 18R_1, R_3 - 12R_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

v_1 leading
 v_2, v_3 free

$$v_1 + v_3 = 0 \quad v_1 = -v_3 \quad \ker(A - I)^2 = \text{span} \left\{ \underbrace{\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}}_{\vec{v}}, \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{\vec{w}} \right\}$$

$$v = \begin{bmatrix} -v_3 \\ v_2 \\ v_3 \end{bmatrix} = v_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + v_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Choose $v^{(2)} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \in \ker(A - I)^2$ (we can also choose any of $\vec{v}, \vec{w}$)
 $\notin \ker(A - I)$

$$(A - I)v^{(2)} = v^{(1)}$$

$$\begin{bmatrix} 4 & 0 & 4 \\ -6 & 0 & -1 \\ -6 & 0 & -6 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} \quad v^{(1)} = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}$$

$$P = \begin{bmatrix} v^{(3)} & v^{(4)} & v^{(2)} \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = A \begin{bmatrix} v^{(3)} & v^{(2)} & v^{(2)} \end{bmatrix}$$

Dependent
P

$$P = \begin{bmatrix} -4 & 0 & -1 \\ -9 & 1 & 1 \\ 6 & 0 & 1 \end{bmatrix}$$

$$e^{At} = P \cdot e^{Jt} \cdot P^{-1}$$

$$e^{Jt} = e$$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} t$$

$$= \begin{bmatrix} e^{-t} & 0 & 0 \\ 0 & e^t & t e^t \\ 0 & 0 & e^t \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & -1 \\ -9 & 5 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} e^{-t} & 0 & 0 \\ 0 & e^t & t e^t \\ 0 & 0 & e^t \end{bmatrix} \cdot P^{-1}$$

Ex Solve $x' = Ax$ where $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\lambda = 1$ $\dim(\lambda) = 3$
 $m(\lambda) = 2$

$$(A-I) = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$v_1 + v_3 = 0$$

$$v_1 = -v_3$$

$$v = \begin{bmatrix} -v_3 \\ v_2 \\ v_3 \end{bmatrix}$$

$$= v_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + v_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\ker(A-I) = \text{span} \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Find $v^{(3)}$ is spanned by $v^{(1)}$ or $v^{(2)}$

$$(A-I)v^{(3)} = v^{(1)}$$

$$(A-I)v^{(4)} = 0$$

$$\text{or } (A-I)v^{(3)} = v^{(2)}$$

$$(A-I)v^{(2)} = 0$$

Choose $(A-I)v^{(2)} = 0$ $(A-I)^2 v^{(3)} = 0$

$$(A-I)^2 = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\ker(A-I)^2 = \mathbb{R}^3 = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Choose $v^{(3)} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$$(A-I)v^{(3)} = v^{(2)}$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \in \ker(A-I)$$

Find $v^{(4)}$

$$v^{(4)} = (A-I)v^{(2)} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ no vector}$$

Then, we must choose $v^{(4)}$ such that it must be in $\ker(A-I)$ and be linearly independent from $v^{(1)}$ and $v^{(2)}$

Choose $v^{(4)} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$. Then.

$$\begin{bmatrix} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} & \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \end{bmatrix} = P$$

$v^{(1)} \quad v^{(2)} \quad v^{(3)}$

$$J = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution: $x_{\text{general}} = P \cdot e^{Jt} \cdot C$

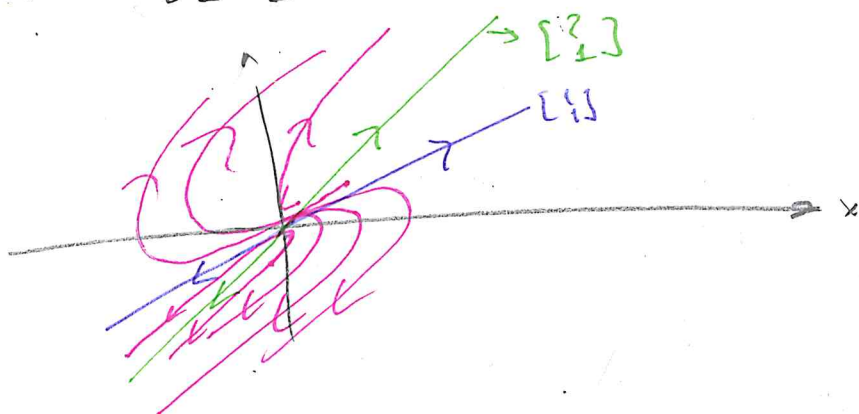
$$= \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e^t & 0 & 0 \\ 0 & e^t & t e^t \\ 0 & 0 & e^t \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$= \begin{bmatrix} -e^t & e^t & t e^t \\ 0 & 0 & e^t \\ e^t & 0 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$= c_1 e^t \underbrace{\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}}_{v^{(1)}} + c_2 e^t \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{v^{(2)}} + c_3 e^t \left(t \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{v^{(2)}} + \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{v^{(3)}} \right)$$

ordinary Comp free generalized term

Ex 7. $P = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ $J = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow$ Draw phase portrait.



$$x_p = c_1 e^t \underbrace{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}_{\text{dominant } t \rightarrow \infty} + c_2 e^{1t} \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\text{dominant } t \rightarrow -\infty}$$

$$t \rightarrow \infty \quad x(t) \rightarrow \infty$$

$$t \rightarrow -\infty \quad x(t) \rightarrow 0$$

(0,0) unstable node

HW
①

Draw phase portraits of
 $P = \begin{bmatrix} 5 & 0 \\ 2 & 1 \end{bmatrix}$ $J = \begin{bmatrix} 4 & 0 \\ 0 & -1 \end{bmatrix}$

② $P = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$ $J = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$

③ $P = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$ $J = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$

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